

## Notes on this revision

Ms. Ref. No. JSPI-D-26-00452 — “WAGER: Within-cell Antisymmetric Gain Evaluation of Resolution”

This note is not a point-by-point response to the referee reports: at the time of preparing this revision, the author did not have the text of the individual reviewer comments in hand. Rather than guess at what was said, this document summarizes the revision comprehensively — what was added, what was corrected, and why — so the editors and reviewers can see everything that changed regardless of which points it does or does not happen to address. The author would welcome the opportunity to prepare a conventional point-by-point response once the reports are available, if the editors judge that necessary before a decision.

### 1. New theoretical results

1. **Bregman-score generalization** (Theorem 2, Section 3.3). The covariance identity previously stated only for the quadratic score is shown to hold for the entire family of scores generated by a strictly convex, differentiable function on the simplex, with the quadratic and log-score identities recovered as named corollaries rather than treated as an unexplained special case and an informal sensitivity check.
2. **Asymptotic normality** (Theorem 3, Section 3.9). A central limit theorem for the dataset-level estimator, under a bounded score and a condition ruling out one cluster from dominating the sample, together with consistency of the already-implemented sandwich variance estimator. This is the formal justification for reading the image-clustered intervals reported throughout the paper as asymptotically valid.
3. **Sensitivity bound for an unrecorded confounder** (Proposition 3 and its worst-case Corollary, Section 3.4–3.5). A Cauchy–Schwarz argument, in the spirit of Rosenbaum (2002) and Cinelli & Hazlett (2020), that bounds how far an unrecorded covariate could move the reported alignment estimate from what a fully adjusted audit would find, given a single elicitable correlation parameter, with a data-free worst-case fallback needing only the label cardinality and score bound.
4. **Relation to Diebold–Mariano/Giacomini–White** (Corollary, Section 3.9). At a trivial audit feature (no declared prior structure), the paper’s own statistic and its interval are shown to reduce exactly to a clustered form of the classical comparative-forecast-evaluation test, situating the method precisely relative to that literature rather than only by analogy.

### 2. New empirical domain

A third application, long-tailed text classification on 20 Newsgroups (Section 4.9), was added alongside the existing scene-graph-generation and long-tailed image classification studies. This domain has neither a spatial nor a class-frequency prior, directly testing the estimator’s claimed generality on a domain unlike either of the first two rather than only asserting it. The two domains’ training-recipe compositions turn out to differ — the same recipe carries the real instance-alignment gain in one domain and almost none in the other — and the revision states this as suggestive pending the fuller (multi-seed, calibration-matched) protocol already applied to the image-classification domain, not as an already-settled finding.

### 3. Corrections identified during further internal review

Beyond the additions above, a second internal review pass after the theoretical and empirical additions were drafted found and corrected the following:

1. **A related-work argument was found to be incorrect and was removed.** Section 2.5 had argued that subtracting two separately-estimated single-model score resolutions does not recover the paper’s own estimate because the relevant finite-sample bias “scales with each model’s own cell counts.” On inspection this is false when both compared models share the same declared audit partition: the same attenuation factor then applies to each model’s own resolution estimate, and subtracting the two recovers the paper’s own estimate of the contrast exactly. The argument was replaced with the paper’s actually-valid distinguishing point — that the classical literature supplies no inference for the covariance between two such estimates from the same data, which this paper’s construction targets directly — and three directly relevant references were added: Ferro & Fricker (2012, *QJRMS*), Siegert (2014, *QJRMS*), and DeLong, DeLong & Clarke-Pearson (1988, *Biometrics*).
2. **The new asymptotic normality theorem’s proof required an explicit condition that was missing.** The original proof sketch invoked a within-cell law of large numbers under a condition that bounds only the aggregate identified sample fraction, not any individual audit cell’s size. An explicit condition was added — the audit feature takes values in a fixed, finite set with positive limiting mass per cell, so every cell’s size diverges in the idealized large-sample limit — and the proof was corrected to use it. The manuscript now states plainly that this is an asymptotic idealization the real scene-graph application has not attained in-sample (most audit cells there hold only the minimum of two examples), and that the reported empirical interval-coverage simulation, not the theorem directly, is the finite-sample evidence for the interval’s practical validity.
3. **The headline empirical claim was restated to reflect a robustness check the paper itself runs.** The audit of a released scene-graph debiasing method (Section 4.6) reports the same comparison under two proper scores. The quadratic score’s calibration-matched interval crosses zero; the log score’s does not (a fact already computed but not previously carried into the manuscript’s stated conclusion or its own table). The revision now states precisely what the two scores agree on — that the method’s improvement is overwhelmingly attributable to fitting the prior rather than to individual-instance evidence — and what they disagree on — whether any instance-alignment remainder is exactly zero — rather than reporting only the reading that happened to show a null result.

#### 4. Reproducibility

All code, cached model outputs, and scripts needed to reproduce every number, table, and figure in the revised manuscript are available at <https://github.com/vinhqdang/WAGER>, including a script that checks every reported value against the committed results files.